ZOO506-Applied-Entomology-And-Pest-Management

SUBJECTIVE&OBJECTIVES

Final-Term File 2023 By

SULMAN ALI (BS-ZOOLOGY)

(0341-7547044)



CURRENT OBJECTIVES

1. The… …extends from the mouth to the anus at the hinder end____***Alimentary canal***
2. It passes over the stomach or other parts of the_____ ***alimentary canal***
3. Adults have hypognathous ***(directed downward)*** mouthparts and unique piercing– sucking larval jaws composed of the mandibles ***(directed downward)*** plus maxillae Insects benefits
4. Mouth of neuropteran_____ ***Downward***
5. How many types of Artificial control of insects?_____ ***6***
6. Types of insect control_____***TWO***
7. _____entirely absent in some insects__***Wings***
8. ***Two*** Pairs of wing present in insect.
9. Fleas are present all the united state aspect ***Alaska.***
10. Insects are present? ***In water, Land: Both***
11. Those insects which nearly kill the other insects? ***Predator***
12. Lice is predator or ***parasite***.?
13. Insect which in used mainly in biological research? ***Fruit fly***
14. IPM stand for? ***Integrated Pest Management***
15. ***Shellac*** is a commercial product.
16. Climate include..... ***humidity***

17. At one place on the surface is a minute opening or group of openings through the shell, called the ***micro Pyle.***

18. Commercial Products is ***Shellac.***

19. Commercial products are except 1-Silk 2-Honey 3-Beeswax ***4-Sugar***

20. Measures for the artificial control of insects may be divided into _____groups: ***Six***

21. Climate in Natural control include _____***Humidity***

22. Those of flies are called ***maggots***

23. Sometimes hairs from the body of the larva, held together by silk, form ***most of the cocoon***

24. The ***parasites*** that live in or on the different stages of other insects, then called hosts

25. The ***predators*** that capture and devour other insects

26. Insects which undergo complete Metamorphosis are called _____. ***Holometabola***

27. Insects have body segments_____***3***

28. Bombyx mori is the scientific name of ***Silk***

29. .....is insect-borne disease. Malaria, bubonic plague,typhiod, typhus, yellow fever ***all***

30. Parasites of many species attack.... ***Insects***

31. Insect control is of.....***2 types***

32. Paurometabola means..... ***Metamorphosis***

33. The ***alimentary canal*** extends from the mouth through about the center of the body to the anus at the hinder end.

34. Habitat of Diplopoda is.... ***on Land***

35. .....is a crustacean living on land***. Watery***

36. Small diplopods are sometime called as..... ***Wireworm***

37. Spines, hairs, scales or other structures are often present on the..... to entirely conceal its surface and its sutures. ***Margin of maxilla***

38. Numerous.....are present on the tips and outer internal margins of the maxillae. ***Spines and hair***

39. The organism in which a parasite live is called.... ***host***

40. Organism that capture other insects called..... ***predator***

41. Honey is a product..... ***Commercial***

42. Natural control includes all except.....***Chemical control***

43. Artificial control of insects includes all except,.....***hand picking***

44. Artificial control of insects may be divided into..... ***6 groups .***

45. Natural control fall those various factors which affect the distribution and ***welfare of insects.***

46. IPM stands for..... ***Integrated pest management***

47. Pesticide resistance in insect should be managed carefully using strategies such application of pesticides with different modes of..... ***anti-resistance strategies***

48. Insects are responsible for a number of commercial products.... ***A third group of insects***

49. All are the commercial products of insects except.... .***sugar***

50. Bombyx mori is the scientific name of..... ***Silk worm***

51. Dry-toasted cricket taste like… . Seed__ ***sunflower seed***

52. Apitherapy more commonly known as… .. Venom therapy___ ***bee venom***

53. insects can be found in… … environment on earth___ ***every***
54. All insects fertilize the soil with the nutrient from their… _____ ***droppings***
55. insect are crucial component of many….whereas they perform many important function_____ ***ecosystems***
56. Maggot only eat… .. And prevent infection naturally_____ ***dead flesh***
57. Fly larva….. were used to treat wounds to prevent or stop gangrene_____ ***maggots***
58. Most silk is produced from… . Of the silkworm, Bombed mari_____ ***cocoon***
59. …..have long been used in genetic studies ***Fruit flies***
60. scales insects provide dyes that are used in many product including…____ ***All of these cakes,medicines cosmetics***
61. …….in the holometabola the feeding periods or instar in referred _____ comes first, followed by a molt and growth____***Adult***
62. the….serves as a distribution food from the place where it is digested to all the part that need it_____***Blood***
63. …….is a crustacean living on land __***obster, crab, beach flea, sowbug***
64. Sense organ of insects are present on. .. ***Head***
65. parasites of many species attack… . Insects___ ***One or few insects***
66. In insects the antennae present in ***Head*** region
67. Near the base of the labium a tongue like structure covered by bristles or hairs which is called ***hypopharynx***
68. Tiny antenna like structure consisting of from one to six (usually five) segments, which is called the ***maxillary palpus***
69. a pair of jaws present behind the labrum is called ***Mandible***
70. The …. are slight grooves or scratches on the surface forming the movable joints _____***sutures***
71. In insects cuticle is present outside the…___***Hypodermis***
72. Adult insects have .... segments. _____ ***7..to..15***
73. The number of ocelli in insects is _____***1-3***
74. The size of ocelli is equal to the one facet of eye. _____***Compound eye***
75. The compound eyes are also called …..eyes. _____ ***ocelli***
76. In adult insects, number of segments in thorax region is…_____***3***
77. The front lip is called _____***Lebrum***
78. The hinder lip is called _____ ***Lebium***
79. There are segments in the abdomen of the adult insect. _____ ***3-11***
80. The ***Crustacea*** are mainly water-inhabiting animals which breathe either by gills or, in the smaller forms, through the surface of the body.
81. ***millipede***, refers to the large number of legs possessed by these animals
82. The Arachnida generally have the segments of the body grouped into ***two*** sections
83. ***Eight(8)*** legs are all attached to the cephalothorax
84. ***mites*** there is no evident division of the body into sections .
85. A pair of antennae is present on the ***head***
86. In insects the fragments of the body are grouped in ***3*** section:
87. Wings present in insects***: 4***
88. In embryo, segments in head ***6 to 3 in thorax, 11 to 12 in abdomen***

89. The cuticula hardens quickly after its production and consists quite largely, at least, of a nitrogenous substance called ***chitin***

90. These sutures and the joint lines divide the cuticula into areas called ***plates or Sclerites***

91. Its inner surface, forming the roof of the mouth, is often called the ***epipharynx***

92. Thorax has segments ***3***

93. The first abdominal segment, when seemingly more apart of the thorax than of the abdomen, is called the ***median segment or ropodeum***

94. In insect the pairs of legs ***3***

95. ***Forelegs*** Use for capillary

96. The jumping power of the grasshopper is due to the great development of its ***hind legs***

97. The ***compound eyes*** are also called ***ocelli*** eyes.

98. Near the base of the labium a tongue like structure covered by bristles or hairs which is called ***hypopharynx***

99. Behind each mandible at the side of the mouth is the ***Labrum.***

100. The ***Sternum*** is the ventral portion of a segment of an arthropod.

101. The front lip is called ***Lebrum***

102. The hinder lip is called ***Lebium***

103. In ***prognathous*** heads ,labrium is present above the mouth opening.

104. Which of the following type of the eye has group of pyramids like structures called ***compound eye***

105. The mandibles and ***maxillae*** move sideways.

106. Thus forming the movable joints; along the other ***infoldings*** it hardens (becomes sclerotized) and becomes rigid and such places show on the surface only as slight grooves or scratches called ***sutures.***

CURRENT SUBJECTIVE

Define disease with an example:

Any harmful deviation from the normal structural or functional state of an organism is called Disease.

Example: the Yellow Fever and the Malaria is the best examples of the disease.

Commercial products of insects name:

A third group of insects is responsible for a number of commercial products

1= Silk 2= Honey 3= Beeswax 4= Shellac 5= Red Dye 6= Flu Vaccines

Two of the most important domesticated insects are the silkworm (Lepidoptera) and the honeybee (Hymenoptera). Some coarse silks are produced from the cocoons of large wild silkworm species. Most commercial silks, however, come from the silkworm Bombyx mori.

Natural control of insects:

Natural control refers to the maintenance of insect populations within certain bounds by environmental conditions, or factors. Both non-biological and biological factors contribute to the natural control of insects. These include: Physical factors such as weather (e.g. cold winters can reduce some insect populations). Natural enemies (predators, parasites, pathogens) can also be used to control insects population:

What is physical control?:

Physical control refers to mechanical or hand controls where the pest is actually attacked and destroyed. Physical controls are used mostly in weed control. Tillage, fire, removal by hand, grazing and mowing are all used to destroy weeds and prevent reproduction.

Methods included:

1= Heat 2= Cold 3= Moisture 4= Light 5= Electricity.

Discuss artificial control Methods of insects:

Measures for the artificial control of insects may be divided into six groups:

(1) Mechanical control: Mechanical pest control is the management and control of pests using physical means such as fences, barriers or electronic wires.

(2) Cultural control: Cultural methods of pest management include use of resistant varieties, tillage, mulching, hand weeding and hoeing, pruning, trapping and hand picking of insects and weeds, and the use of physical barriers such as row covers and sticky bands.

(3) Biological control: Biological control is the use of living organisms to suppress pest populations, making them less damaging than they would otherwise be.

(4) Legislative control: It is the control of insect pest by enforcing insect laws by the government to prohibit the spread and multiplication of insect pests in the agricultural fields.

(5) Physical control: Physical control refers to mechanical or hand controls where the pest is actually attacked and destroyed. Physical controls are used mostly in weed control.

(6) Chemical control: Chemical control is using pesticides, fungicides and bactericides to control pests and diseases.

Natural control of insects:

Natural control refers to the maintenance of insect populations within certain bounds by environmental conditions, or factors. Both non-biological and biological factors contribute to the natural control of insects. These include:

- Physical factors such as weather (e.g. cold winters can reduce some insect populations).
- Quality and quantity of available food.
- Competition between species or among individuals of the same species.
- Amount and quality of living space or territory.
- Natural enemies such as predators and parasites.

How insects attack on injurious species:

Those insects which attack injurious species probably form the most important group. They are divided into two sections:

1-The parasites that live in or on the different stages of other insects, then called hosts

2-The predators that capture and devour other insects.

Parasites usually, though not always, live in or on a single host insect during their entire development. Whereas a single predator may derive its nourishment from many victim insects. Parasites are of many species, frequently so specialized in their habits that one kind of parasite attacks only one or a few host species In some cases the parasites work only in insect eggs and thus prevent any feeding loss by the host insect. Other parasites attack early stages of the pests, in some instances killing them soon.

Benefits of insects in details:

Insects pollinate many of our fruits, flowers, and vegetables. We would not have much of the produce that we enjoy and rely on without the pollinating services of insects, not to mention honey, beeswax, silk, and other useful products that insects provide.

Insects feed on a seemingly endless array of foods. Many insects are omnivorous, meaning that they can eat a variety of foods including plants, fungi, dead animals, decaying organic matter, and nearly anything they encounter in their environment.

Many insects are predatory or parasitic, either on plants or on other insects or animals, including people. Such insects are important in nature to help keep pest populations (insects or weeds) at a tolerable level.

Insects are underappreciated for their role in the food web. They are the sole food source for many amphibians, reptiles, birds, and mammals. They are a rich source of protein, vitamins, and minerals, and are prized as delicacies in many third-world countries.

Name of different insect legs:

1= Coxa 2= trochanter (two in a few cases) 3= femur 4= tibia 5= tarsus

Role of insects in pollination:

- Pollination is the transfer of pollen grains to the ovary of the plant.
- Pollination by insects is called entomophily.
- Pollination by insects happens when insects go in search of food. Flowers produce a sugary liquid called nectar which many insects consume on a large basis. When insects land on a flower, pollen grains tend to stick to their bodies.
- The insect then moves from one to another flower of the same species, pollen gets transferred to the stigma of flowers and hence causing pollination.
- Therefore pollination by insects occurs when insects move from one plant to another in search of food.

What is the Coxa:

The coxa is the segment that articulates with the body, frequently partly lying in a more or less cup shaped hollow of the latter. It may be short or long and is generally freely movable on the body and powerful.

What is the pollination:

The process of transfer of pollen grains from anther to stigma belonging to the same flower or different flowers is called pollination. The two types of pollination are self-pollination and cross-pollination.

Enlist the plants which require pollination:

Strawberries, apples, raspberries as well as broad beans, peas and courgettes are just a few of the edible plants that rely on visiting insects to take pollen from flower to flower – a process known as pollination. This then leads to fertilization and fruit formation.

Physical Transmission:

In case of mechanical transmission the pathogen is simply carried externally or internally by the insects. Mostly Caused insects that have biting/chewing mouthparts.

How allergy is caused?

Allergy (hay fever, asthma, etc.) are frequently caused by exposure of susceptible persons to dried insect scales and hairs.